Laser system for identification, tracking, and control of flying insects

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Abstract: Flying insects are common vectors for transmission of pathogens and inflict significant harm to humans and agricultural production in many parts of the world. We present proof of principle for an optical system capable of highly specific vector control. This system utilizes a combination of optical sources, detectors, and sophisticated software to search, detect, and identify flying insects in real-time, with the capability of eradication using a lethal laser pulse. We present data on two insect species to show species distinction; *Diaphorina citri,* a vector of the causal agent of citrus greening disease, and *Anopheles stephensi,* a malaria vector.

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OCIS codes: Fourier optics and signal processing; (100.0100) Image processing; (110.0110) Imaging systems; (120.0120) Instrumentation, measurement, and metrology; (150.0150) Machine vision.

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#260161 Received 15 Mar 2016; revised 14 Apr 2016; accepted 25 Apr 2016; published 23 May 2016
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Corrected: 24 May 2016

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1. Introduction

Control and monitoring of disease-vector insects is critical to global health. Insect vectors have a role in spreading pathogens amongst humans, animals, and agricultural products. Such pathogens create worldwide strain on healthcare and food resources. Current techniques for insect control create undesirable impacts as they are often based on chemical or biological agents.

Because of the global economic and health impacts of pathogen-carrying insects, both monitoring and control are needed. Insect monitoring currently uses traps or long distance radar. Traps are limited in their coverage area and require human intervention to replace and record, which may be expensive and unreliable. The most common form of insect-monitoring radar, LIDAR [1], uses reflected light to detect insects. It is expensive to implement, and is best employed from a far range, such as from a flying craft [2], and so far offers no control options. Some of the recent identification development in LIDAR is quite promising, but so far works only on large insects and sometimes relies on labeling [3, 4]. Using reflected light, LIDAR works in a fundamentally different mode from the presented system. Other forms of optical detection are being developed, but are limited by either tight physical constraints, a reliance of tagging insects, or identification ability [5–8]. The purpose of control is to somehow alter a population, often by eradication of individuals, to limit the negative impact of the population. Here we propose eradication of individuals to limit access to an area. Control is mostly done broadly with chemical pesticides, which have non-target effects on organisms such as fish and birds [9]. We present a new modality of insect monitoring which has a mid-range operating distance, real-time identification using wing beat frequency, and highly specific control.

In this work, we demonstrate the ability of a new system to monitor and control two important insect vectors. These choices are *Anopheles stephensi* mosquitoes (Diptera: Culicidae) and *Diaphorina citri* psyllids (Hemiptera: Psyllidae). There are several motivations for this choice. *D. citri* are relatively small (3-4 mm) flying insects which disperse well and can readily infiltrate horticultural facilities [10, 11]. This vector has a significant global economic impact in citrus production since they transmit *Candidatus Liberibacter asiaticus*, the pathogen which causes citrus greening disease [12, 13]. Currently used abatement techniques are limited in effectiveness [14]. There are technical considerations as the small

#260161 Received 15 Mar 2016; revised 14 Apr 2016; accepted 25 Apr 2016; published 23 May 2016
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size and speed of this vector poses a demanding test of an optical tracking system. *An. stephensi* have a narrow body with a wing span of about 5.5 to 6.8 mm [15] and are interesting targets because of their role vectoring malaria, which results in 200 million infections and 200,000 deaths annually [16].

The Photonic Fence (PF) is a system which implements monitoring and control by advancing previous mid-range insect tracking and identification technologies [17–19] through complex integration to a stage where field deployment is practical [20]. A photograph of the system is shown in Fig. 1(a) and is diagrammed in Fig. 1(b), depicting a combination of light sources and detectors which scans a defined boundary, and tracks and identifies flying insects as they cross this boundary. In a configuration not shown, PF also has the ability to deliver a lethal source of energy onto those insects whose identification profile is within a database. The ultimate choice of a control laser is dependent on many factors which affect mortality. This was closely tested on mosquitoes in a previous study [21].

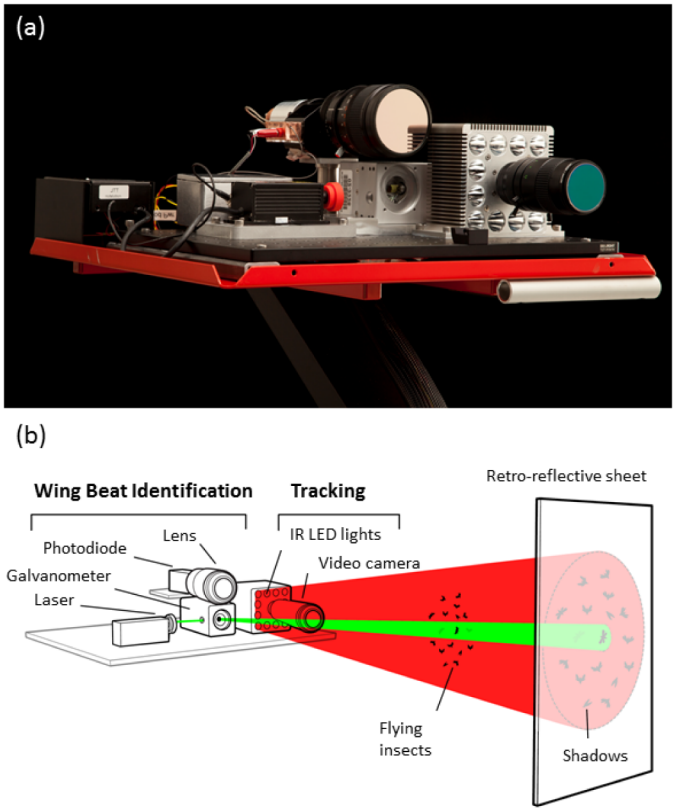


Fig. 1. (a) A photograph of the PF tracking and identification module. The compact unit consists of an illumination source, a camera, and the tracking laser. The eradication module also attaches to the unit, but is not shown for clarity. (b) A diagram depicting the system level operation of the PF unit. Flying insects crossing a virtual boundary are tracked and identified through measurement of wing beat frequency.

PF is comprised of independent modules. There is a motion detector tracking module, an identification module which uses a laser to track and measure wing beat frequency as the signature for identification, and an eradication module that can deliver a lethal dose appropriate for the vector. For the data presented here, the unit is optimized to have a maximum range of 30 m, which usually exceeds practical constraints imposed by local geographical obstructions in expected use cases. For the testing, we used a distance of 8 m. This system allows for specificity and accuracy of control. Although it would not be suitable for control on a massive scale, it is ideal for targeted control of insects entering a sensitive

#260161 Received 15 Mar 2016; revised 14 Apr 2016; accepted 25 Apr 2016; published 23 May 2016
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area, such as an opening to a green house. It can be used either for close range survey or for supplementary control.

PF operates through sophisticated software which integrates the module components and provides real-time coverage. The system analyzes images in front of the retro-reflective strip that is placed at an operator-defined distance from the PF platform. This establishes the range of the unit. The system continuously images a volume in front of the retro-reflector. It forms a virtual boundary within which flying insects are interrogated. In the camera image, the volume appears two dimensional with the retroreflector in the background. The virtual boundary is formed by only imaging a subset of the available volume. As detailed below, spatial positioning and tracking are achieved through real-time reconstruction of the silhouettes of the insects, while identification is achieved through an optical lock-in technique whereby the wing beat frequency of the insect is measured.

Coverage for this system may be tailored to the specific environment, and a range of roughly 8 m is optimal to allow for both effective control and concurrently maintain power consumption sufficiently low so as to allow for battery operation. Although the modules may be configured to operate well beyond this range, limitations set by the cost and power requirements would then result in compromised practicality.

2. Methods

2.1 Testing configuration

Tests of the flight tracking and wing beat detection capabilities used approximately 200 *D. citri* placed in a 45 L glass aquarium with a retro-reflector (model 983-10NL, 3M) placed behind the aquarium, at a distance of 8 m from the PF unit (Fig. 1). The imaging area was restricted to the tank, behind which was the retro-reflector. The tank was 0.318 m tall by 0.533 m wide and 0.267 m deep. There was a total of 8 m between the camera and the retro-reflector. The illuminating source consists of 12 low-power (1 W) light emitting diode operating in the infrared (850 nm) pointing at the retroreflector (Fig. 1). The imaging camera (model a-741, PixelLink) output was running at 103 frames per second with a maximal integration time lasting from one frame to another. It was transmitted via IEEE 1394 interface to a Dell Precision M6500 laptop computer, which runs custom software utilizing CUDA libraries in conjunction with an nVidia graphics card. For wing beat identification, the targets were then illuminated with a green (532 nm) laser which is steered by a galvanometer (SCANcube 7, ScanLab) (Fig. 1). The laser was rated to 100 mW but filtered down to 5mW continuous wave (CW) with neutral density filters. To maximize tracking accuracy during wing beat measurement, the tracking camera ran at a higher frame rate with a smaller field of view: 736 frames per second at a resolution of 176 x 128 pixels. A lens and photodiode assembly help detect the retro-reflected light. The photodiode signal is sampled at 1 MHz and transmitted to the laptop computer via a Universal Software Radio Peripheral 2 (USRP 2).

The detection module operated for 11 hours and the wing beat identification system was actively collecting data for a total of 24 min, during which time it recorded 835 *D. citri* flight segments of 10 ms or longer. Some segments lasted as long as 113 ms, with the median segment lasting 20 ms. Based on visually observed behavior and the length of observation, we estimate that at least a majority of individuals were recorded. The signal was processed by a two-point finite difference filter [22] to remove constant offsets. This method is used as a high pass filter to remove constant or slowly varying offsets in the detected laser power. High frequency noise was removed with a second order Butterworth low pass digital filter [23] with a corner frequency of 2 kHz, and the filtered signal was compressed to conserve storage space. The Welch overlapped periodogram method [24] was used to estimate the average power spectrum of the wing beat signal.

At the USDA Subtropical Agricultural Research Station [25], local *D. citri* were collected. *D. citri* were collected from local highly populated citrus trees and visually identified. The temperature was held at 28 °C with the humidity ranging from 40% to 90%.

#260161 Received 15 Mar 2016; revised 14 Apr 2016; accepted 25 Apr 2016; published 23 May 2016
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An. stephensi mosquitoes were cultured in an insectary held at 24 °C and 75% humidity. Testing and data collection were taken in the same set-up as described above. All of the testing for the mosquito occurred in Bellevue, Washington.

2.2 High-speed videography

High speed videography was used to quantify the fundamental wing beat frequency. This more direct measure helped us validate the wing beat frequency measured by our system.

The high speed videography analysis used recordings of flying *D. citri* obtained using a digital high-speed video camera (model Color Sensor Phantom V12.1, Vision Research Inc.) at frame rates ranging from 1000 fps to 11000 fps, with resolutions maximized to saturate the available 6.4 gigapixels per second of bandwidth.

The frames are generated too fast to be saved and referred to later, so a simple triggering system was built to only save instances in which activity likely occurred. The insect enclosure used for high speed videography contained a triggering system consisting of two laser-photodiode pairs, positioned such that the two laser beams intersect at a large angle, with photodiodes opposite each laser to monitor the intensity of the laser light. Any *D. citri* flying within a small region around the intersection of the beams reduced the laser intensity read by the photodiodes and triggered camera data recording. This triggered saving of the video segment immediately before and after the trigger time.

2.3 Lethality tests

Lethality tests on *D. citri* show susceptibility to laser radiation. The test described was performed with anesthetized insects using a CW 445 nm laser diode with 670 ± 10 mW (95% confidence) power. The laser diode was driven by a current-regulated diode driver (model FlexMod P3, Nautil Integration). The beam was shaped with a focusing lens and two perpendicular cylindrical lenses (Fig. 2). It was attenuated with a neutral density filter. It had an elliptical profile (0.50 mm x 0.39 mm) and no divergence 5 mm above and below the plane of dosing. The driver circuit was controlled by a digital pulse generator (model Arduino Pro Mini, Sparkfun Electronics). The microcontroller was programmed to generate on command a single pulse of configurable duration, with duration specified to a precision of 0.1 ms and entered via a serial port connection. The error in pulse lengths was measured using a 100 MHz oscilloscope (model 196C, Fluke Corporation) and found to have ± 0.05% mean error.

In the lethality experiments, anesthetized *D. citri* were systematically exposed to laser pulses of variable duration. Prior to the start of each experiment, *D. citri* were placed in a clear acrylic box and anesthetized with CO_2. Each *D. citri* was dosed in sequence by the laser and allowed to recover in air overnight. The dose was applied on the side of the thorax and in the same plane relative to the laser. To control for mortality due to the anesthetization procedure, unexposed control specimens were anaesthetized in the same manner and number as those in the treatment group, but left to sit in the test apparatus with no laser exposure. Insects exposed to a specific energy level were pooled for the purposes of estimating the population survival rate [26].

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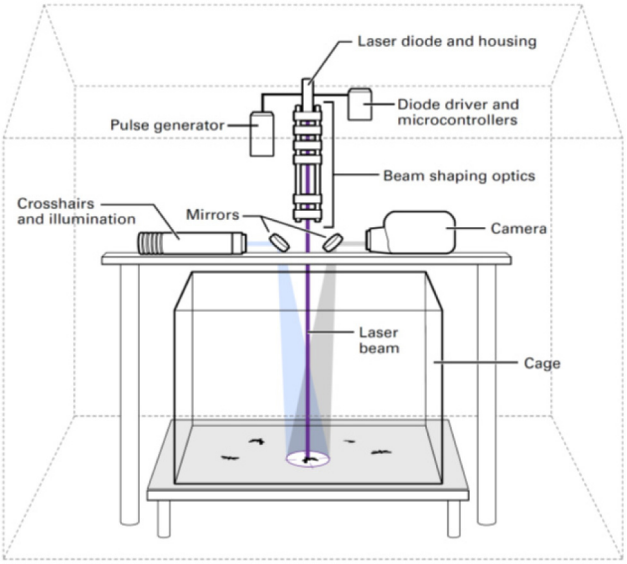


Fig. 2. The lethality test apparatus is shown. The eradication laser was tested in an operator-controlled environment for accurate lethality measurements. Survival is assessed using WHO guidelines for pesticides as described in the text.

3. Results and technique

At the system level, PF functions sequentially, where initial detection is achieved through video image acquisition of flying insects crossing a virtual boundary. The image is analyzed to detect motion. The insects whose speeds fall within a user-defined range are tracked and then interrogated by a green laser which helps identify by measuring the wing beat frequency. By comparison to a stored database, those insects found to be harmful individual vectors may then be eradicated by a laser pulse. Three frames acquired every 10 ms are needed for tracking. Even though a minimum of 30 ms is needed to identify a path for tracking, the software tracks for 200 ms before proceeding to the wing beat identification step. It does this by comparing every 10 frames. This adds assurance that the time spent by the wing beat identification system, which can only do one at a time, is well spent. For wing beat identification, five sequential wing beats are needed. Since *D. citri* show a wing beat of 187 Hz, this calculates to a minimum of 27 ms. This ability of rapid analysis of a large data set is key to achieving a high level of reliability in tracking and identification. We describe the details of the system and its capabilities below.

3.1 Motion detection and measurement

The PF tracking function monitors an area by illuminating a retro-reflector using an infrared light source. Flying insects traversing this space are back-illuminated by light coming back from the retro-reflector and cast silhouettes, which are imaged by a camera and analyzed in real-time. It is the silhouettes and not shadows that are imaged. In the tested situation, the focal plane of the camera is well within the cage dimensions. For a system to make use of the full volume, at least half of the distance between the camera and the retroreflector is covered by the focal distance of this set-up. Although there are 12 tightly spaced sources of light for the illumination system, the silhouettes present one usable image for the camera. The minimum required image contrast for acquisition is fundamentally set by the intensity of the reflected infrared light as measured by the imaging camera. Custom software forms an intensity pattern of the captured image. Objects that traverse the light field result in low intensity values on the occluded pixels, and the pixelated image is converted into binary with

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an adjustable threshold. Using a historical differencing engine and connected component labeling [27], a centroid is calculated and defines the spatial center of the object.

The module detects motion by comparing the current video frame to a recent frame with a user-defined delay time, as diagrammed in Fig. 3 (top). A positive value in the pixel intensity difference between frames is a signature for motion, as it indicates a region that currently contains an insect but did not during a previous time period. The analysis software also has configurable size thresholds to reject large unrelated objects in the field of view. The delay time between analyzed frames is 10 ms. Only 3 analyzed frames are needed for tracking. To reduce false positive signals, a total of 200 ms of continuous tracking was a requirement for proceeding on to wing beat analysis. Flexibility in these times allows for optimization of the system based on the characteristic speed of the insects of interest in the deployed area. The difficulty of this task and the effectiveness of this procedure are visually apparent in the time-sequenced frames shown in Fig. 3 (bottom). Here, the motion of the insects is barely perceptible to the eye against the patterned stationary background of the retro-reflector. However, the tracking module easily identifies the insects within the field of view. Spatial positioning is obtained by assigning each tracked object a coordinate in image space based on its centroid, from which the actual azimuth and elevation relative to the optical center of the camera are calculated.

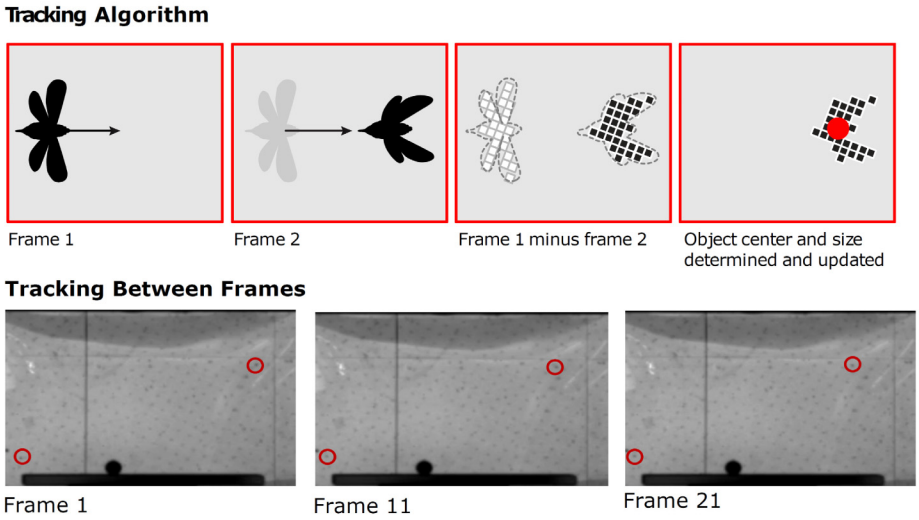


Fig. 3. The algorithm for tracking is shown (Top). The tracking module detects flying insects through motion, through subtraction of time-sequenced digitized frame. Also shown are the real images (bottom). In this sequence, the motion of the insects (in red) is barely perceptible to the eye, while the tracking algorithm performs this task efficiently. The collected video relating to the frames below is in Visualization 1.

The detection module was tested for 11 hours, as detailed in Methods. The median observed speed was 0.064 radians per second and 95% of observed velocities were between 0.010 and 0.15 radians per second, which is well within the module's tracking capability. The camera acquisition time is fast enough such that the retro-reflector could be as narrow as several centimeters to capture an insect in flight. Instantaneous speeds were approximated by dividing distance traveled between successive frames by the frame period. The overwhelming majority (99%) of recorded flight segments of *D. citri* were sufficiently slow to operate within the tracking window of the system.

3.2 Identification

The motion detection module provides position information for the identification module, which then determines insect type through wing beat frequency measurements. The targets are

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illuminated with a green (532 nm) laser steered by a galvanometer using position information from the detection module. As the insect beats its wings, the photodiode detector measures the oscillating light intensity, providing information about the frequency at which the size of the silhouette oscillates and thus allowing for identification. Stray light sources are rejected by modulating the green laser at a 25 kHz carrier frequency, and using this frequency as a reference for lock-in amplification detection [28] to distinguish the signal from background. A video demonstration of PF's tracking an insect, while being interrogated by the green laser is shown in movie Visualization 1.

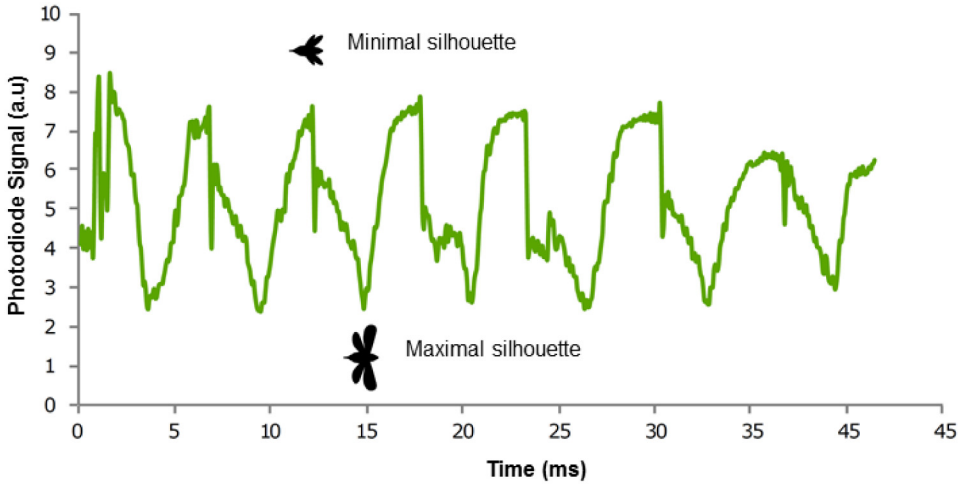


Fig. 4. A photodiode signal of the wingbeat used for identification of insects. The intensity pattern varies temporally. The signal increases as occlusion decreases.

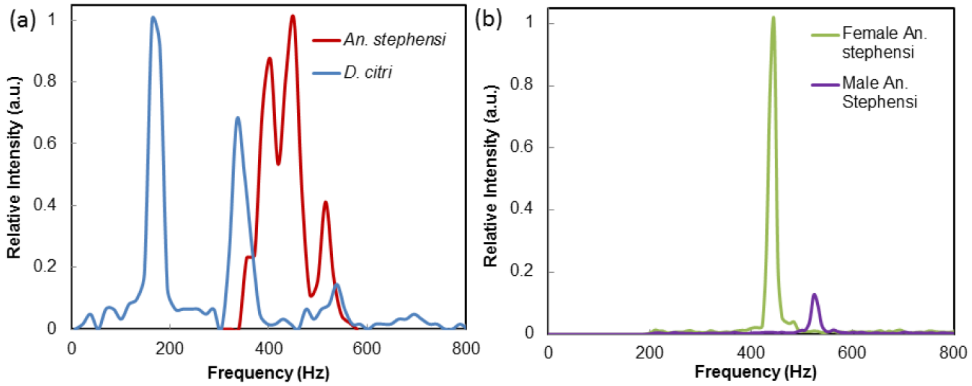


Fig. 5. (a) The spectral signatures of *D. citri* and *An. stephensi.* The fundamental frequencies are distinct, with multiple harmonic peaks shown in the *D. citri* spectra. (b) Spectral signatures for a male and female individual *An. stephensi.* The entire spectra are shown in Fig. 7.

Figure 4 shows a typical photodiode signal from a tracked *D. citri*, demonstrating the variation of the intensity pattern with time as the insect beats its wings. Higher values indicate greater brightness, so peaks relate to apparent minimum size of the insect and troughs relate to when the insect appears at its maximum size. Insect wing beats do not occlude light in a perfect sinusoidal pattern, and the waveform contains both harmonics and noise. The frequency resolution is sufficient to distinguish the fundamental and harmonic modes, leading to highly specific signatures for identification. Figure 5 shows transformed wing beat spectra with the DC component from the laser in Fig. 5 digitally filtered out. In Fig. 5(a), the spectral signatures of individual groups of *D. citri* and *An. stephensi* are plotted concurrently,

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demonstrating clear differences and enabling identification. Figure 5(b) shows individual spectra for one male and one female *An. stephensi* mosquitoes. This precise capability is critical to enabling a disruptive method of control which is highly effective and with minimal environmental impact.

The wing beat frequency measurements for *D. citri* were verified through high speed videography. The video analysis consisted of wing beat periods obtained from 50 video segments featuring 2 to 154 wing beat periods, and periods were measured by counting elapsed frames between equivalent wing positions. The distribution featured a mean frequency of 187 ± 26 Hz (95% confidence), providing validation of the PF data.

3.3 Eradication

The eradication module delivers a pulse of lethal energy to those insects whose wing beat frequency matches the profile of the vector to be controlled. The control laser is steered in the same manner as the tracking laser, and delivers a dose that is controlled by either the dwell time or power. Although this function was not integrated for the in-flight testing, mortality of such a laser on *D. citri* is below. Laser-induced mortality was more generally evaluated on mosquitoes for different laser types and settings in a previous study [21].

For accurate lethality and dosage tests, the PF control laser was tested in isolation from the rest of the system in an operator-controlled environment, shown in Fig. 2. In this apparatus, the laser delivers a controlled pulse of variable duration at constant output intensity, with an average power density of 440 W/cm^2. The pulse times range from 3 ms to 28 ms (± 0.1 ms), and survival is assessed using World Health Organization guidelines for pesticide evaluation [26].

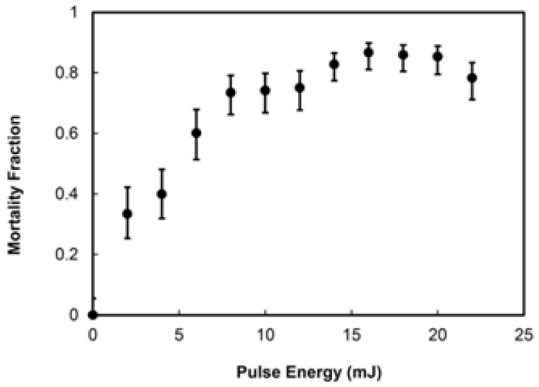


Fig. 6. Mortality of *D. citri*. The insects were exposed to different energy levels through variation of pulse time. The wavelength of the eradication laser is 445 nm at a power level of 670 mW. LD:50 occurs at approximately 7 mJ.

A significant, dose-dependent change in mortality is observed among psyllids, shown in Fig. 6. Dosing measurements for *D. citri* show the onset of mortality occurs at a low energy level of 2 mJ, and 90% mortality is reached at 15 mJ after the data was shifted such that the mortality of no dose controls corresponds to zero. This observation that mortality can be achieved at such low levels of laser energy is crucial for PF field deployment, as power energy requirements are a significant factor for cost-effective and stand-alone operation.

4. Discussion

Several issues were of concern during development, including tracking accuracy of insects possessing high flight speeds. Although *D. citri*'s small size and fast dispersal makes it a challenging application choice for this technology, the results never-the-less demonstrate the proficiency of the tracking module. This performance gives us confidence in the approach. Further, the design of the system is flexible and allows for a wide window of detection in

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insect speed, through configuration of the retro-reflector width or image acquisition times. Several PF units may be integrated to provide coverage to larger areas. As a new technique, PF's identification through measurement of wing beat frequency required validation. Accuracy of the measured wing beat frequency was validated through high speed videography. Although it can sometimes be the case that the harmonics are more intense than the fundamental, the fundamental was still easily identified. This is shown in Fig. 7.

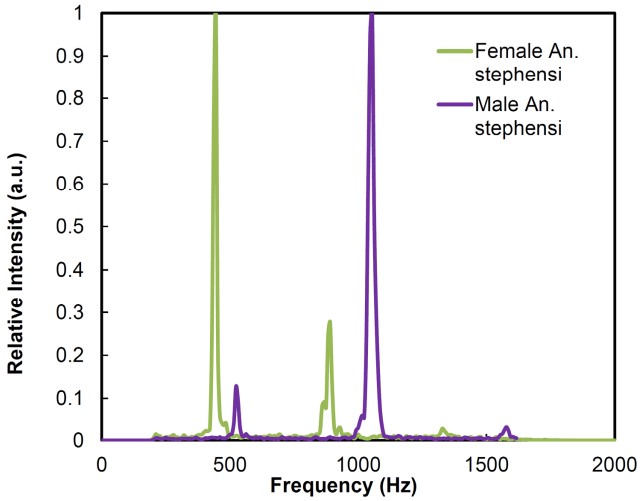


Fig. 7. Full spectra.

Dosing studies presented here for *D. citri* show efficacy at low energies and the lethal dose may be further optimized through selection of laser wavelength. We believe the eradication mechanism is thermal, in that it leads to a rise in the body temperature of the insects [29]. The relatively low power required to deliver a lethal dose is fortuitous and critical, as it enables nearly autonomous operation.

We believe PF presents a new potential modality of monitoring and control of insects. A particularly useful case would involve a small number of insects moving into a sensitive area for which current abatement techniques are not effective. An example of this case is with young crops vulnerable to citrus greening disease. Grove borders or entries to nurseries present this particularly vulnerable area in which additional investment in monitoring and control could be worth additional investment. Control is difficult to achieve and the primary strategy for mitigation is replacement of the infected trees, which is extraordinarily expensive.

PF's ability to track and differentiate insects using wing beat frequency provides agricultural producers and researchers a new modality for monitoring movement and location of insects. This is a significant advancement over current methods which rely on traps treated with attractants for population studies. As a stand-alone unit, PF requires no human intervention, with low operating cost as error and latency are reduced.

Specific identification of insect species and gender would require a database built using collected environmental data. With such a database, it would be possible to make predictions about the insect using their wing beat frequency [30]. This would involve thoroughly documenting the wing beat frequency in the field of different *An. stephensi* genders. We recognize that species and gender classification by fundamental wing beat frequency is not trivial as even within some mosquito species there is overlap [5]. However, a wing beat identification device could be used with several other already collected data as well as environmental information for identification. For example, in a nursery for orange trees, one would expect to see *D. citri* but not *An. stephensi*.

#260161 Received 15 Mar 2016; revised 14 Apr 2016; accepted 25 Apr 2016; published 23 May 2016
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Since this system is self-contained and controlled by software, custom safety and environmental factors can be added. This technology already has low potential to yield non-target damage due to the specific identification preceding a lethal dose of individual insects. To add an additional layer of safety, expanding the virtual boundary on the sides of the retro-reflector field of view would allow for a programmed emergency shut off if a large object suddenly entered the field, such as a worker. If the system was installed such that a tree branch was present in the field of view, that area could similarly be cut off from the target area. Ideally, the system would be set-up such that the field of view is not restricted, especially horizontally. This includes protecting it from transitory large objects which may block the field of view, such as farm animals. Other environmental factors which affect the insect wing beat frequency, such as temperature, could be accounted for with additional studies, but it not in the scope of this work. We recommend further studies on other insects and environments if a particular use case is identified. As such, we are optimistic that with further development, PF could become a new method of study and control for other insect vectors.

Acknowledgments

The authors thank Bill and Melinda Gates for their support and sponsorship through the Global Good Fund. Additional funding for this study was provided by the USDA Agricultural Research Service. We are grateful to A. Gomezplata, O. Zamora, D. Stockton, and G. Betts for their assistance in this study.

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